

DEMO 1 SIH :

Start date :18/08/2026

End date :20/08/2026

(actually its 24h hackathon, but considering our classes try to finish it by 20 or 21st!!!)

PROPOSED PROBLEM STATEMENT AND SOLN :

Core idea: An intelligent grievance platform where citizens report problems in plain language, and the system automatically understands, prioritizes, routes, tracks, and verifies resolution — not just a complaint box.

Problems it solves: Citizens not knowing which department to contact, misrouted complaints, no transparency, indefinite pending status, duplicate reports scattered across people, minor complaints burying urgent ones, no escalation, and authorities lacking visibility into recurring issues.

How it works (workflow):

1. **Citizen submits** a grievance in natural language (with photo/location optional) — no need to know which department handles it.
2. **System auto-classifies** it — category, subcategory, location, severity, and the right department.
3. **Duplicate detection** — groups similar complaints (e.g., 17 people reporting the same pothole) into one "master grievance," showing real impact.
4. **Priority engine** — ranks urgency using severity, number of people affected, how long it's persisted, location sensitivity (school/hospital), and repeat complaints — not just submission order.
5. **Auto department assignment** — routes to the correct authority (roads, electrical, water, sanitation, etc.).
6. **Authority dashboard** — officials see grievances sorted by priority/status instead of a random queue.
7. **SLA tracking** — each complaint type has an expected resolution time, actively monitored.
8. **Auto-escalation** — unresolved cases move up the chain (officer → senior → dept head) instead of sitting idle.
9. **Citizen-facing transparency** — a visible timeline showing status, department, and progress.
10. **Resolution verification** — citizen confirms if the issue was *actually* fixed before it's closed (prevents fake "resolved" tags).
11. **Feedback loop** — citizens rate resolution quality and time taken.
12. **Analytics** — spots patterns like recurring problem areas or underperforming departments.
13. **Hotspot mapping** — visualizes complaint clusters geographically to flag problem zones.
14. **Proactive shift** — aims to move from "citizen complains → govt reacts" to "system flags emerging pattern → govt acts early."

15. **Multilingual support** — accepts regional languages (e.g., Malayalam) so citizens don't need to translate first.

8 Core Modules:

1. Citizen Grievance Portal
2. Intelligent Grievance Processing (classification)
3. Duplicate & Related Complaint Detection
4. Priority Engine
5. Authority Management
6. SLA & Escalation Engine
7. Resolution & Citizen Verification
8. Governance Analytics

WHAT MAKES THIS SOLUTION INNOVATIVE:

1. Turning scattered complaints into collective evidence

Typical system: 17 people file 17 separate complaints about the same pothole. Each sits in a queue, ranked by submission time. Authority sees "17 tickets" — looks like 17 minor issues.

Our system: It detects these describe the same underlying problem and merges them into one master grievance: *"Road damage at ABC Junction — 17 citizens affected."* That single number changes how the authority perceives urgency. One complaint looks ignorable; 17 linked citizens on the same spot looks like a mandate to act. This is the difference between noise and signal.

2. Priority based on real-world impact, not submission order

Typical system: First-come-first-served. A streetlight complaint filed Monday gets handled before a dangerous pothole near a school reported Wednesday, purely because of timing.

Our system: Priority is computed from severity + number of people affected + duration + location sensitivity (school/hospital nearby) + repeat frequency. So the school pothole automatically jumps ahead of the streetlight issue — even though it arrived later — because the system understands *consequence*, not just chronology.

3. Escalation that doesn't rely on humans remembering

Typical system: If an officer ignores a complaint, it just stays "pending" indefinitely. There's no built-in accountability mechanism.

Our system: SLA timers run in the background. Miss the deadline → automatic reminder → still nothing → auto-escalate to senior officer → then department head. The complaint physically cannot get "lost" in someone's inbox. This directly targets the "delays" problem most existing portals never solve — they add a submission form but no enforcement layer.

4. Verifying resolution from the citizen's side, not the officer's

Typical system: Officer clicks "Resolved." Ticket closes. Nobody checks if the pothole was actually filled.

Our system: Marking something "resolved" triggers a citizen confirmation step. If the citizen says no, it reopens or escalates automatically instead of quietly dying as a closed ticket. This closes a loophole that's extremely common in real government portals — inflated "resolution rates" that don't reflect reality.

5. Reactive → proactive governance

Typical system: Waits for complaints to accumulate, then someone eventually notices a pattern manually (if ever).

Our system: Because every complaint is geo-tagged and categorized, the system can flag emerging clusters itself: "*40% rise in water-related complaints in Ward 6 over 3 weeks — potential pipeline issue.*" Government acts on a trend before it becomes 500 individual complaints. This shifts the model from "fix what broke" to "notice what's breaking."

6. Accessibility through language, not literacy in bureaucracy

Typical system: Citizen has to know both the right department name and often has to write in English/formal language.

Our system: A citizen writes in Malayalam, casually, without knowing what department handles roads. The system does the translation *and* the classification. This matters a lot in India specifically — most citizens who face real infrastructure problems (rural roads, water issues) are the least likely to navigate an English-only bureaucratic form.